

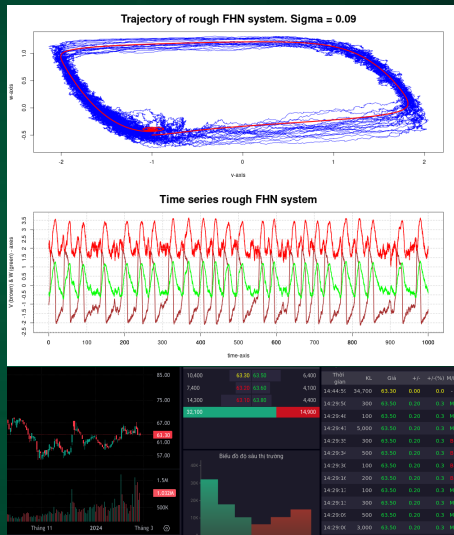
Nonautonomous dynamics via rough paths: theory and applications

Duc H. Luu

(MPI MIS & IMH-VAST)

Research interests

- Stochastic processes, random/nonautonomous dynamical systems, rough path theory, neural (controlled) differential equations
- Learning the stable dynamics with non-Lipschitz and chaotic behaviors: numerical attractors
- Learning the non-autonomous dynamics: trend detection.
- Signature theory for data streams: order book data (finance)
- Topological space of chemical substances (chemistry).



Outline

Rough path theory

Nonautonomous dynamics and numerics

Applications and open questions

Motivation from stochastic calculus

- path-wise RDEs on $(\Omega, \mathcal{F}, \mathbb{P}, (\theta_t)_{t \in R})$: Sell, Oseledets, Millionschikov, Kloeden, Arnold...

$$\dot{y} = f(\theta_t \omega, y). \quad (1.1)$$

- SDEs with Brownian motions: Itô & Stratonovich calculus. Arnold, Kloeden, Caraballo, Schmalfuss, Crauel, Scheutzow...

$$dy_t = f(y_t)dt + g(y_t)dW_t. \quad (1.2)$$

- SDEs with fractional Brownian motions: rough differential equations. **Rough path theory** (Lyons 1998), Gubinelli, Friz, Nualart, Hairer...

$$dy_t = f(y_t)dt + g(y_t)dB_t^H, H \in (\frac{1}{4}, 1). \quad (1.3)$$

- Solution Itô-Lyons map: $IL(\omega) = y(T, \omega, y_0)$ of random ODEs and SDEs is **measurable** w.r.t. ω . But $IL(\omega)$ is **continuous** w.r.t. ω (Lyons' universal theorem).
- **Question 1: How to define the stochastic integrals in (1.2) and (1.3) ?**
- **Question 2: how to interpret equation $dy = f(y)dx$ given a driving path x ?**

- fractional Brownian motion (fBm) with Hurst parameter $H \in (0, 1)$: a continuous centered Gaussian process $B^H = (B^H(t), t \in \mathbb{R})$, with covariance function

$$R(s, t) = \frac{1}{2}(|t|^{2H} + |s|^{2H} - |t - s|^{2H}), \quad s, t \in \mathbb{R},$$

When $H = 1/2$, B is the standard Brownian motion.

- B^H is neither Markov nor semi-martingale if $H \neq \frac{1}{2}$!
- Kolmogorov-Centsov theorem: B_t^H is H^- - Hölder continuity a.s. in t .
- For simplicity, think of $dy = g(y)dx$ with scalar path x . First solve the ODE $\dot{u} = g(u)$ to obtain solution $\pi(t, y_0)$ and then assign $y_t = \pi(x_t, y_0)$. Then

$$dy_t = \frac{\partial \pi}{\partial x}(x_t, y_0) dx_t = g(\pi(x_t, y_0)) dx_t = g(y_t) dx_t.$$

By Taylor expansion

$$\begin{aligned} y_{a,b} &= \frac{\partial \pi}{\partial x}(x_a, y_0) x_{a,b} + \frac{1}{2} \frac{\partial^2 \pi}{\partial x^2}(x_a, y_0) (x_{a,b})^2 + \dots \\ &= g(\pi(x_a, y_0)) x_{a,b} + \frac{1}{2} \frac{\partial}{\partial x} g(\pi(x_a, y_0)) (x_{a,b})^2 + \dots \\ &= g(y_a) x_{a,b} + \frac{1}{2} Dg(y_a) g(y_a) (x_{a,b})^2 + \dots \end{aligned}$$

One needs more information from higher order terms if x is of low regularity.

Rough paths

In the simplest form for $\beta \in (\frac{1}{3}, \frac{1}{2})$, a couple $\mathbf{x} = (x, \mathbb{X})$, with $x \in C^\beta([a, b], \mathbb{R}^d)$ and $\mathbb{X} \in C_2^{2\beta}([a, b]^2, \mathbb{R}^d \otimes \mathbb{R}^d)$ is called a *rough path* if they satisfy Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u} \otimes x_{u,t}, \quad \forall a \leq s \leq u \leq t \leq b. \quad (1.4)$$

We write $\mathbb{X}_{s,t} = (x \otimes x)_{s,t} =: \int_s^t x_{s,u} \otimes dx_u$ and call it **Lévy area** of x .

- $X_t = \int_0^t a_s dB_s$, we define the stochastic integrals $\int y dX$ as the integral w.r.t. the local martingale X .

$$\mathbb{X}_{s,t} = \int_s^t X_{s,u} dX_u = \frac{1}{2} X_{s,t}^2 - \frac{1}{2} (\langle X \rangle_t - \langle X \rangle_s).$$

- When $X = B^H$ is a fractional Brownian motion, we define the stochastic integral $\int y \delta B^H$ in the sense of Skorohod-Wick-Itô by using the Wick product. Then we can compute explicitly

$$\mathbb{X}_{s,t} := \int_s^t B_{s,u}^H \delta B_u^H = \frac{1}{2} (B_{s,t}^H)^2 - \frac{1}{2} (t^{2H} - s^{2H}).$$

Rough integrals

- $\beta > \frac{1}{2}$: consider $y \in C^\beta([a, b], V \otimes H)$ and $x \in C^\beta([a, b], H)$. Thanks to **the sewing lemma**, there exists a Young integral $\int_a^b y dx$ defined by

$$\int_a^b y dx = \lim_{|\Pi| \rightarrow 0} \sum_{[s,t] \in \Pi} y_s x_{s,t}. \text{ Moreover,}$$

$$\left\| \int_s^t y_u dx_u - y_s x_{s,t} \right\| \leq K_\beta |t - s|^{2\beta} \|y\|_{\beta, [a,b]} \|x\|_{\beta, [a,b]}.$$

- $\frac{1}{2} \geq \beta > \frac{1}{3}$: $y \in C^\beta([a, b], V \otimes H)$ is **controlled by** $x \in C^\beta([a, b], H)$, i.e. $\exists y' \in C^\beta([a, b], V \otimes H \otimes H)$, $R^y \in C^{2\beta}(\Delta^2([a, b]), V \otimes H)$ st

$$y_{s,t} = y'_s x_{s,t} + R^y_{s,t}, \quad \forall \min I \leq s \leq t \leq \max I.$$

Due to **the sewing lemma**, the rough integral $\int_s^t y_u dx_u$ can be defined (**Gubinelli (2004)**) as $\int_s^t y_u dx_u := \lim_{|\Pi| \rightarrow 0} \sum_{[u,v] \in \Pi} [y_u x_{u,v} + y'_u \mathbb{X}_{u,v}]$.

Moreover

$$\begin{aligned} & \left\| \int_s^t y_u dx_u - y_s x_{s,t} - y'_s \mathbb{X}_{s,t} \right\| \\ & \leq C_\beta |t - s|^{3\beta} \left(\|x\|_{\beta, [s,t]} \|R^y\|_{2\beta, \Delta^2[s,t]} + \|y'\|_{\beta, [s,t]} \|\mathbb{X}\|_{2\beta, \Delta^2[s,t]} \right). \end{aligned}$$

Existence and uniqueness theorem: RDEs

- Finite dimension: RDE $dy_t = f(y_t)dt + g(y_t)dx_t$ is understood in the integral form

$$y_t = y_0 + \int_0^t f(y_s)ds + \int_0^t g(y_s)dx_s. \quad (1.5)$$

- **Riedel - Scheutzow (2017)**: solve the solution in the **Friz-Victoir sense**.
- **LHD (2022)**: solve in the **Gubinelli sense**, i.e. solution of the form (y, y') with $y, y' \in C^\beta$, $y' = g(y)$.
- (H_f) $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is locally Lipschitz continuous and of one-sided linear growth

$$\exists C > 0 : \langle y, f(y) \rangle \leq C(1 + \|y\|^2), \quad \forall y \in \mathbb{R}^d; \quad (1.6)$$

in addition f is of linear growth in the perpendicular direction, i.e. there exists $C_f > 0$ such that

$$\left\| f(y) - \frac{\langle f(y), y \rangle}{\|y\|^2} y \right\| \leq C_f (1 + \|y\|), \quad \forall y \neq 0; \quad (1.7)$$

- (H_g) either $g(y) = Cy + g(0)$ or $g \in C_b^3(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^m, \mathbb{R}^d))$ such that

$$C_g := \max \left\{ \|g\|_\infty, \|Dg\|_\infty, \|D_g^2\|_\infty, \|D_g^3\|_\infty \right\} < \infty; \quad (1.8)$$

- **Method: Doss-Sussmann technique and stopping time analysis.**

Rough difference equations

Discrete time set $\Pi = \{t_k\}_{k \in \mathbb{N}}$ defined by

$$y_{t_0} \in \mathbb{R}^d, \quad y_{t_{k+1}} = y_{t_k} + f(y_{t_k})(t_{k+1} - t_k) + F_{t_k, t_{k+1}}(y, \mathbf{x}), \quad (2.1)$$

$$F_{s,t}(y, \mathbf{x}) = \begin{cases} g(y_s)x_{s,t}, & \mathbf{x} = x \\ g(y_s)x_{s,t} + Dg(y_s)g(y_s)\mathbb{X}_{s,t}, & \mathbf{x} = (x, \mathbb{X}) \end{cases} \quad (2.2)$$

(H_f) $f \in C(\mathbb{R}^d, \mathbb{R}^d)$ of linear growth, i.e.

$$\exists C_f > 0 : \quad \|f(y)\| \leq C_f \|y\| + \|f(0)\|, \quad \forall y \in \mathbb{R}^d; \quad (2.3)$$

(H_g) either $g \in C_b^2(\mathbb{R}^d, \mathbb{R}^{d \times m})$ such that

$$\|g\|_\infty, C_g := \max \left\{ \|Dg\|_\infty, \|D^2g\|_\infty \right\} < \infty. \quad (2.4)$$

or $g(y) = Cy + g(0)$ for some matrix C .

For Brownian motion B , (2.1) is the Milstein scheme.

Challenge: Solution norm estimation for rough differential/difference systems is still an open question when g only Lipschitz continuous. Different from stochastic approach.

Euler scheme and convergence

LHD - Kloeden (2023): Consider the RDE

$$d\hat{y}_t = f(\hat{y}_t)dt + g(\hat{y}_t)dX_t, \hat{y}_a \in \mathbb{R}^d, t \in [a, b] \quad (2.5)$$

where X is a stochastic process with stationary increment which almost all realizations x are of local Hölder continuous with exponent $\nu \in (1/3, 1/2]$. If in addition to $(\mathbf{H}_f)$, $(\mathbf{H}_g)$, f and D^2g are local Lipschitz, then there exists a generic random variable $\hat{\xi}(\hat{y}_a, x, [a, b])$ such that the following estimate holds

$$\begin{aligned} & \sup_{0 \leq k \leq m} \|\hat{y}(t_k, y_a, x) - y_{t_k}\| \\ & \leq \hat{\xi}(\hat{y}_a, x, [a, b]) (|\Pi[a, b]| + \|x\|_{\rho, [a, b], |\Pi[a, b]|}^p)^{3-p} (b - a + \|x\|_{\rho, [a, b]}^p) \\ & \leq C(\omega) |\Pi[a, b]|^{3\nu-1}. \end{aligned} \quad (2.6)$$

In particular,

$$\lim_{|\Pi[a, b]| \rightarrow 0} \sup_{0 \leq k \leq m} \|\hat{y}(t_k, \hat{y}_a, x) - y_{t_k}\| = 0. \quad (2.7)$$

Open question: does $C(\cdot)$ belong to $\mathcal{L}^1$ or $\mathcal{L}^2$? **Still the gap between stochastic and nonautonomous numerics.**

Generation of RDS

- Continuous time: **Bailleul-Riedel-Scheutzow (2017)** proves that RDE $dy_t = f(y_t)dt + g(y_t)dx_t$ generates a continuous RDS $\varphi(t, \omega, y_0)$ on $\mathbb{R}^d$ which is the solution of the RDE w.r.t. the driving path $x(\omega)$ and at initial point y_0 .
- Discrete time: Define the discrete mapping $H^\Delta(x)y = y + f(y)\Delta + F_{0,\Delta}(y, x)$. In **LHD-Kloeden (2023)**, we prove that the discrete Euler scheme (2.1) generates a discrete random dynamical system $\varphi^\Delta : \Pi^\Delta \times \Omega \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ over $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ such that

$$\begin{aligned}\varphi^\Delta(0, \omega)y_0 &= y_0, \\ \varphi^\Delta(k\Delta, \omega)y_0 &:= H^\Delta(\theta_{(k-1)\Delta}\omega) \circ \dots \circ H^\Delta(\omega)y_0, \quad \forall k \geq 1.\end{aligned}\tag{2.8}$$

Throughout this paper, we assume that θ is an ergodic dynamical system. In **LHD-PT.Hong-ND.Cong (2024)**, we prove the ergodicity for fBm B^H .

Existence of pullback attractors

Now we assume further that f is dissipative: there exist $c, d > 0$ so that

$$\langle y, f(y) \rangle \leq c - d\|y\|^2, \quad \forall y \in \mathbb{R}^d. \quad (2.9)$$

LHD (2022): Assume $(\mathbf{H}_f), (\mathbf{H}_g^b), (\mathbf{H}_X)$, then there exists a random pullback attractor $\mathcal{A}$ such that $|\mathcal{A}(\cdot)| \in \mathcal{L}^\rho$ for any $\rho \geq 1$. If in addition f is global Lipschitz continuous then the random attractor is upper semi-continuous with respect to the noise intensity in the sense that $\mathcal{A} \rightarrow \mathcal{A}_0$ a.s. (w.r.t. the Hausdorff semi-distance) as $C_g \rightarrow 0$, both in the almost sure and in $\mathcal{L}^\rho$ senses. Moreover, if f is strictly dissipative then $\mathcal{A}$ is a singleton for $C_g \ll 1$.

LHD-Kloeden (2023), ND. Cong-LHD-PT. Hong (2023) Assume $(\mathbf{H}_f), (\mathbf{H}_g)$ and the dissipativity condition (2.9). Consider system (2.1) with the regular time set Π^Δ , where $\Delta \in (0, 1)$ satisfied the inequality

$$0 < \Delta < 1 \wedge \frac{d}{2C_f^2} \wedge \frac{1}{2d}. \quad (2.10)$$

Then the generated discrete random dynamical system φ^Δ of (2.1) admits a random pullback attractor $\mathcal{A}^\Delta$. Moreover, for C_g small enough, the numeric attractor $\mathcal{A}^\Delta$ converges to the attractor $\mathcal{A}$ in the Hausdorff semi-distance, i.e. $d_H(\mathcal{A}^\Delta | \mathcal{A}) \rightarrow 0$ as $\Delta \rightarrow 0$, a.s.

Fitzhugh-Nagumo neuro-system

The deterministic FHN neuron model is

$$\begin{cases} dv_t &= v_t - \frac{v_t^3}{3} - w_t + I \\ dw_t &= \varepsilon(v_t - \mu w_t + J) \end{cases} \quad (3.1)$$

f is dissipative, but does not satisfy $\left\| f(y) - \frac{\langle f(y), y \rangle}{\|y\|^2} y \right\| \leq C_f (1 + \|y\|)$. We introduce a different Lyapunov function

$$V(y) := \left(1 + v^4 + \frac{6}{\varepsilon \mu} w^2 \right)^{\frac{1}{4}} \Rightarrow \|\nabla V\| \leq L_V. \quad (3.2)$$

LHD-J.Jost (2024): Under the assumptions $(\mathbf{H}_g^b)$, $(\mathbf{H}_X)$, there exist constants $\delta > 0$ such that for any $\lambda \in (0, 1)$ small enough, there exist a constant $K_\lambda > 0$

$$V(y_t) - K_\lambda \leq e^{-\delta t} \left[V(y_0) - K_\lambda \right] + H(\|x\|_{p, [0, t]}), \quad \forall t \in [0, T] \quad (3.3)$$

where $H(\xi) = L_V(16C_p C_g)^p \lambda^{1-p} \xi^p + 8L_V C_p C_g \xi$. As a result, there exists a global pullback attractor for the rough FHN system.

Open problem: Numerical attractor for rough FHN system? **Tamed ?**

Bifurcation

- Pitchfork bifurcation: $dy = (\mu y - y^3)dt + g(y)dB^H$. When $\mu > 0$, $g \equiv g_0$ and $H = \frac{1}{2}$, Crauel proves that additive noise destroy bifurcation, i.e. there exists a unique stationary distribution. We consider $g \in C_b^3$ with $g(0) \neq 0$ and $H > \frac{1}{3}$ and prove the existence of a global random pullback attractor $\mathcal{A}$. Moreover, when $\mu < 0$ then $\mathcal{A}$ is a random singleton. **Open problem**: is $\mathcal{A}$ a singleton when $\mu > 0$?
- Poincaré-Andronov-Hopf bifurcation:

$$dy = \begin{pmatrix} \lambda y_2 + y_1(\mu - y_1^2 - y_2^2) \\ -\lambda y_1 + y_2(\mu - y_1^2 - y_2^2) \end{pmatrix} dt + g(y)dB^H, \quad y = (y_1, y_2)^T \in \mathbb{R}^2.$$

Again, due to dissipativity, we can prove the existence of a global random pullback attractor $\mathcal{A}$, which is a random singleton when $\mu < 0$. **Open problem**: if $\mathcal{A}$ is not a singleton when $\mu > 0$, how can we define a random limit set $\mathcal{A}(\omega)$? How close $\mathcal{A}(\omega)$ to the limit circle Γ of the unperturbed system?

- **Challenge**: in both cases, f is **not globally Lipschitz continuous**, hence semi-continuity of $\mathcal{A}$ w.r.t. g is not guaranteed.

Stochastic gradient descent

- Minimization problem

$$\min_{\theta \in \mathbb{R}^d} Q(\theta) = \min_{\theta \in \mathbb{R}^d} \frac{1}{m} \sum_{i=1}^m Q^{(i)}(\theta), \quad Q^{(i)} \in \mathcal{C}^1. \quad (3.4)$$

When $m \gg 1$, (3.23) is replaced by: at each step t choose a random set Ω_t of $\{1, \dots, m\}$ (with or without replacement) and then updating

$$\theta_{t+1} := \theta_t - \eta \frac{1}{|\Omega_t|} \sum_{i \in \Omega_t} \nabla Q^{(i)}(\theta_t) = \theta_t - \eta \nabla \tilde{Q}_t(\theta_t) \quad (3.5)$$

No matter how random the choice, can rewrite SGD recursion

$$\theta_{t+1} = \theta_t - \eta \nabla Q(\theta_t) + \eta^\alpha \epsilon_t(\theta_t). \quad (3.6)$$

- (3.6) can also be viewed as the Euler scheme of the "limiting equation"

$$d\theta = -\nabla Q(\theta)dt + \sigma(\theta)dX_t, \quad (3.7)$$

where $\sigma\sigma^T = \eta^{2\alpha} \mathbb{E}[\epsilon\epsilon^T | \theta_t = \theta]$.

- If Ω_t i.i.d. then CLT gives $\alpha = \frac{1}{2}$ and X -Gaussian. But not necessary i.i.d.
- **Questions: How does the distribution of (3.6) converge to some limit?**
And which calculus is used to interpret and solve the limiting equation?

Residual Neural Networks via rough paths

- Recurrent neural network: hidden layer equation

$$h_0 = x, \quad h_{t+1} = f(W_t h_t + b_t), \quad y = L(h_{t+1})$$

In the ResNet approach,

$$h_{t+1} = h_t + f(W_t h_t + b_t)\eta. \quad (3.8)$$

- **Chen - Rubanova - Bettencourt - Duvenaud (2018)**: if one iterates continuously with infinitely many layers, one can view (3.8) as a step of the Euler scheme of a controlled ODE - neural differential equation,

$$\dot{h}_t = f_\theta(h_t + b_t), \quad h_0 = x. \quad (3.9)$$

Thus

$$L(h_T) = L(x + \int_0^T f_\theta(h_t + b_t))$$

- **Problem**: for time series data, the scheme is very unstable every time a new data is included, making a jump.

Residual Neural Networks via rough paths

- Kidger - Chen - Lyons (2021): view it as a controlled differential equation

$$h_{t+1} = h_t + f_\theta(h_t + b_t)dX_t \Rightarrow dh_t = f_\theta(h_t + b_t)dX_t \quad (3.10)$$

which is interpreted as a rough differential/difference equation.

- Bayer-Friz-Tapia (2023): Stability of ResNet

$$\begin{aligned} \sup_{k=0..n} \|h_k - \tilde{h}_k\| &\leq C'_{p,n} \exp \left\{ C_{p,n} \|f\|_{C_b^3}^p \left(\|\mathbf{x}\|_{p,[0,n]}^p + \|\tilde{\mathbf{x}}\|_{p,[0,n]}^p \right) \right\} \\ &\quad \times (\|\mathbf{x}_0 - \tilde{\mathbf{x}}_0\| + \rho_p(\mathbf{x}, \tilde{\mathbf{x}})) \end{aligned} \quad (3.11)$$

- **Challenge:** In reality, data might not follow a rough difference equation in the form (3.10). The problem is to find $\ell^* \in \mathbb{R}^{d^*}$ such that

$$\ell^* \in \arg \min_{\ell} L_{\text{price},\alpha}(\ell), \quad \text{where}$$

$$\begin{aligned} L_{\text{price},\alpha}(\ell) &:= \sum_{i=1}^N (S_n(\ell)t_i - S_{t_i})^2 + \alpha(\ell) \\ &= \sum_{i=1}^N \left(S_0 + \ell_0 \langle \mathbf{e}_1, \widehat{\mathbb{X}}_{t_i} \rangle + \sum_{0 < |I| \leq n-1} \ell_I \langle \tilde{\mathbf{e}}_I, \widehat{\mathbb{X}}_{t_i} \rangle - S_{t_i} \right)^2 + \alpha(\ell), \end{aligned}$$

Financial data



References

1. LHD, Jürgen Jost. How rough path lifts affect expected return and volatility: a rough model under transaction cost. SIAM J. Fin. Math., 2023.
2. Nguyen Dinh Cong, LHD, Phan Thanh Hong. Numerical attractors via discrete rough paths. JDDE, 2023.
3. LHD, Peter Kloeden. Numerical attractors for rough differential equations. SIAM Num. Anal., 2023.
4. LHD. Random attractors for dissipative systems with rough noises. DCDS-A, 2022.
5. LHD. Controlled differential equations as rough integrals. PAFA, 2022.
6. Nguyen Dinh Cong, LHD, Phan Thanh Hong. Pullback attractors for stochastic Young differential delay equations. JDDE, 2022.
7. LHD, Phan Thanh Hong, Nguyen Dinh Cong. Stability criteria for rough systems. arxiv 2024.
8. LHD, Jürgen Jost. Attractors of FitzHugh-Nagumo system under stochastic noises: a pathwise approach. (working paper 2024)

THANK YOU FOR YOUR ATTENTION!

Generating metric dynamical system θ

Bailleul-Riedel-Scheutzow (2017): Denote by $T^2(\mathbb{R}^d) = \mathbb{R} \oplus \mathbb{R}^d \oplus \mathbb{R}^{d \otimes 2}$ the truncated step-2 tensor algebra, then $T^2(\mathbb{R}^d)$ is a Banach space with the norm $|\omega| := \max_{k=0,1,2} \|\pi_k(\omega)\|$ where $\|\pi_k(\omega)\|$ is Euclidean norm on $(\mathbb{R}^d)^{\otimes k}$. Define $T_1^2(\mathbb{R}^d) = \{\omega \in T^2(\mathbb{R}^d) : \pi_0(\omega) = 1\}$ together with a $\bullet$ operator

$$(1, g^1, g^2) \bullet (1, h^1, h^2) = (1, g^1 + h^1, g^2 + h^2 + g^1 \otimes h^1), \quad (3.12)$$

for any $\mathbf{g} = (1, g^1, g^2)$, $\mathbf{h} = (1, h^1, h^2)$. Then $(T_1^2(\mathbb{R}^d), \bullet)$ is a group with the identity element $\mathbf{1} = (1, 0, 0)$, and each element $\mathbf{g}$ has its inverse element $\mathbf{g}^{-1} = (1, -g^1, -g^2 - g^1 \otimes g^1)$. In addition, for a rough path $\mathbf{x}$ one can easily check from Chen's relation (3.16) that

$$(1, \mathbf{x}_{s,u}, \mathbb{X}_{s,u}) \bullet (1, \mathbf{x}_{u,t}, \mathbb{X}_{u,t}) = (1, \mathbf{x}_{s,t}, \mathbb{X}_{s,t}), \quad \forall s \leq u \leq t.$$

Also, one can view $\mathbf{x}_t = (1, \mathbf{x}_{0,t}, \mathbb{X}_{0,t})$ as a map from T to $T_1^2(\mathbb{R}^d)$.

Denote by $\mathcal{C}_0^p(T, T_1^2(\mathbb{R}^m))$ the space of all paths $\mathbf{g} : T \rightarrow T_1^2(\mathbb{R}^m)$. Assign $\Omega := \mathcal{C}_0^p(T, T_1^2(\mathbb{R}^m))$ and equip it with the Borel σ -algebra $\mathcal{F}$. Denote by θ the **Wiener-type shift**

$$(\theta_t \omega)_\cdot = \omega_t^{-1} \bullet \omega_{t+\cdot}, \quad \forall t \in T, \omega \in \mathcal{C}_0^p(T, T_1^2(\mathbb{R}^m)), \quad (3.13)$$

and define the so-called *diagonal process* $\mathbf{X} : T \times \Omega \rightarrow T_1^2(\mathbb{R}^m)$, $\mathbf{X}_t(\omega) = \omega_t$ for all $t \in T, \omega \in \Omega$.

Discrete functions and norms

Davie (2008), Lejay (2010) on Euler scheme for RDE.

Bayer-Friz-Tapia (2023): deep neural network stability via discrete rough paths.

Let $[a, b] \subset \mathbb{R}$ and $\Pi = \{a = t_0 < t_1 < \dots < t_n = b\}$ be an arbitrary (finite) set of points of $[a, b]$. We introduce various norms which are natural discrete versions of the continuous ones

$$\|y\|_{\infty, \Pi} := \sup_{t_i \in \Pi} \|y_{t_i}\|;$$

$$\|y\|_{p, \Pi} := \sup_{t_i^* \in \Pi, 0 \leq i \leq r, t_0^* < t_1^* \dots < t_r^*} \left(\sum_{i=0}^{r-1} \|y_{t_i^*} - y_{t_{i+1}^*}\|^p \right)^{1/p};$$

$$\|y\|_{p, \Pi} = \|y_a\| + \|y\|_{p, \Pi};$$

$$\|F\|_{p, \Pi} := \sup_{t_i^* \in \Pi, 0 \leq i \leq r, t_0^* < t_1^* \dots < t_r^*} \left(\sum_{i=0}^{r-1} |F_{t_i^*, t_{i+1}^*}|^p \right)^{1/p},$$

$$\|F\|_{\infty, \Pi} := \sup_{a \leq s \leq t \leq b} \|F_{s, t}\|.$$

For a path ξ we write $\xi_{s, t} = \xi_t - \xi_s$.

Doss-Sussmann transformation

- LHD (2020): the solution $\phi_\cdot(\mathbf{x}, \phi_a)$ of $d\phi_u = g(\phi_u)dx_u$, $u \in [a, b]$, $\phi_a \in \mathbb{R}^d$ is C^1 w.r.t. ϕ_s , and $\frac{\partial \phi}{\partial \phi_a}(\cdot, \mathbf{x}, \phi_a)$ is the solution of the linearized system

$$d\xi_u = Dg(\phi_u(\mathbf{x}, \phi_s))\xi_u dx_u, \quad u \in [a, b], \xi_a = Id, \quad (3.14)$$

- Cass-Litterer-Lyons (2013): **Greedy times**

$$\tau_0 = \min I, \quad \tau_{i+1} := \inf \left\{ t > \tau_i : \|\mathbf{x}\|_{p, [\tau_i, t]} = \gamma \right\} \wedge \max I. \text{ Assign } N(\gamma, \mathbf{x}, I) := \sup \{ i \in \mathbb{N} : \tau_i \leq \max I \}.$$

- The Doss-Sussmann transformation $y_t = \phi_t(\mathbf{x}, z_t)$ there is an one-to-one correspondence between a solution y_t of (2.5) on a certain interval $[0, \tau]$ and a solution z_t of the associate ordinary differential equation

$$\dot{z}_t = \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t) \right]^{-1} f(\phi_t(\mathbf{x}, z_t)), \quad t \in [0, \tau], \quad z_0 = y_0. \quad (3.15)$$

Discrete rough paths

- A non negative function ω defined on $\Delta\Pi := \{(s, t) \in \Pi^2 | s \leq t\}$ is called a *discrete control function* on Π if it vanishes on the diagonal, i.e. $\omega_{s,s} = 0, \forall s \in \Pi$, and is superadditive, i.e. for all $s \leq u \leq t$ in Π

$$\omega_{s,u} + \omega_{u,t} \leq \omega_{s,t}.$$

- For given $[a, b]$ and Π a (finite) discrete time set in $[a, b]$ as above, let $x : \Pi \rightarrow \mathbb{R}^m$ and $\mathbb{X}(\cdot, \cdot) : \Pi^2 \rightarrow \mathbb{R}^{m \times m}$ be discrete functions defined on Π and Π^2 respectively. One then says that x can be lifted to a discrete rough path $\mathbf{x} = (x, \mathbb{X})$ if it satisfies Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u} \otimes x_{u,t}, \quad s \leq u \leq t; \quad s, u, t \in \Pi. \quad (3.16)$$

One can then furnish $\mathbf{x}$ with a semi-norm

$$\|\mathbf{x}\|_{p, \Pi[a,b]} = \left(\|x\|_{p, \Pi[a,b]}^p + \|\mathbb{X}\|_{p/2, \Pi[a,b]}^{p/2} \right)^{1/p}. \quad (3.17)$$

- Example:

$$\mathbb{X}_{t_k, t_l}^{\mu, \nu} := \sum_{j=k}^{l-1} x_{t_k, t_j}^{(\mu)} x_{t_j, t_{j+1}}^{(\nu)}.$$

Sewing lemma

Let $\Pi = \{t_i : 0 \leq i \leq n, a = t_0 < t_1 < \dots < t_n = b\}$ be a finite partition of $[a, b]$ and F be an function defined on $\Delta\Pi$, vanished on the diagonal, i.e. $F_{s,s} = 0, \forall s \in \Pi$. Put

$$\begin{aligned}(\delta F)_{s,u,t} &:= F_{s,t} - F_{s,u} - F_{u,t}, \quad s \leq u \leq t \in \Pi, \\ I_{k,l} &:= \sum_{k \leq j \leq l-1} F_{t_j, t_{j+1}} - F_{t_k, t_l}, \quad t_k \leq t_l \in \Pi.\end{aligned}$$

Assume that for a discrete control ω on Π and a number $\lambda > 1$ we have for all $s \leq u \leq t$ in Π the following inequality

$$\|(\delta F)_{s,u,t}\| \leq \omega_{s,t}^\lambda. \quad (3.18)$$

Then there exists a constant $K > 0$ depending only on λ such that

$$\|I_{k,l}\| \leq K \omega_{t_k, t_l}^\lambda, \quad \forall t_k \leq t_l \in \Pi. \quad (3.19)$$

Ergodicity of θ

- Garrido-Atienza, Schmalfuss (2011): the Wiener shift $\eta_t x = x_{t+} - x_t$ is ergodic on $\Omega' = C^{0,\beta}(\mathbb{R}, \mathbb{R}^m)$ w.r.t. $\mathbb{P}_H$.
- Friz-Hairer (2014, Theorem 10.4): there exists a full measure subset $\Omega_1 \subset \Omega'$ such that $\omega = (1, x, \mathbb{X}) \in \Omega$ for any $x \in \Omega_1$. Moreover, by Friz-Victoir (2010, Theorem 15.42, 45), one can choose this full measure subset Ω_1 such that it satisfies the piecewise linear approximations (mollifier approximation).
- Then consider the natural lift $\mathcal{S}$ on smooth paths

$$\mathcal{S}(x)_{s,t} = (1, x_{s,t}, \int_s^t x_{s,r} dx_r), \quad (3.20)$$

which can be extended to Ω_1 such that $\mathcal{S} : \Omega_1 \rightarrow \Omega$. By Bailleul, Riedel, Scheutzow (2017), there exists a metric dynamical system $(\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathbb{P}}, \theta)$ such that $\hat{\Omega} \subset \Omega$ and $\hat{\mathbb{P}} = \mathbb{P}^H \circ \mathcal{S}^{-1}$. Furthermore,

$$\theta_t \mathcal{S}(x) = \lim_{n \rightarrow \infty} \hat{\theta} \mathcal{S}(x^{(n)}) = \lim_{n \rightarrow \infty} \mathcal{S}(\eta_t x^{(n)}) = \mathcal{S}(\eta_t x). \quad (3.21)$$

- Since η is ergodic, by Garrido-Atienza, Schmalfuss (2011, Lemma 3), θ is also ergodic.
- From now on, we assume additionally that $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ is ergodic.

Gradient decent method

- Minimization problem

$$\min_{\theta \in \mathbb{R}^d} Q(\theta), \quad Q \in C^1. \quad (3.22)$$

- Global solution: $\theta^* \in \arg \min_{\theta \in \mathbb{R}^d} Q(\theta) \Rightarrow \nabla Q(\theta^*) = 0$. This leads to an approximation scheme to achieve θ^* , which is the zero solution of $\nabla Q(\theta) = 0$

$$\theta_{t+1} = \theta_t - \eta \nabla Q(\theta_t), \quad (3.23)$$

for the learning rate $\eta > 0$. (3.23) is called Gradient descent method.

- When η small enough, (3.23) can also be viewed as the Euler scheme for solving the ODE

$$\dot{\theta} = -\nabla Q(\theta), \quad (3.24)$$

where one attempts to solve for the steady state θ^* of (3.24).

Pullback attractors

Roughly speaking, an invariant random compact set $\mathcal{A} \in \mathcal{D}$ is called a *pullback attractor* in $\mathcal{D}$, if $\mathcal{A}$ attracts any closed random set $\hat{D} \in \mathcal{D}$ in the pullback sense, i.e.

$$\lim_{t \rightarrow \infty} d_H(\varphi(t, \theta_{-t}\omega) \hat{D}(\theta_{-t}\omega) | \mathcal{A}(\omega)) = 0, \quad (3.25)$$

where $d_H(\cdot | \cdot)$ is the Hausdorff semi-distance, i.e.

$d_H(D | A) := \sup_{d \in D} \inf_{a \in A} \|d - a\|$. A random set $\mathcal{B} \in \mathcal{D}$ is called *pullback absorbing* in the universe $\mathcal{D}$ if $\mathcal{B}$ absorbs all closed random sets in $\mathcal{D}$, i.e. for any closed random set $\hat{D} \in \mathcal{D}$, there exists a time $t_0 = t_0(\omega, \hat{D})$ such that

$$\varphi(t, \theta_{-t}\omega) \hat{D}(\theta_{-t}\omega) \subset \mathcal{B}(\omega), \text{ for all } t \geq t_0. \quad (3.26)$$

$$\mathcal{A}(\omega) = \bigcap_{t \geq 0} \overline{\bigcup_{s \geq t} \varphi(s, \theta_{-s}\omega) \mathcal{B}(\theta_{-s}\omega)}. \quad (3.27)$$

Signatures of rough paths

Let $x \in C^{1-var}(I, \mathbb{R}^d)$. x_t^i is the coordinate i of x_t at time t . Define

$$\mathbf{g}^{k; i_1, \dots, i_k} := \int_s^t \int_s^{u_k} \dots \int_s^{u_2} dx_{u_1}^{i_1} \dots dx_{u_k}^{i_k}.$$

and the set

$$\mathbf{g} = \left(\mathbf{g}^{k; i_1, \dots, i_k} : 1 \leq k \leq N; i_1, \dots, i_k \in \{1, \dots, d\} \right)$$

to be the N -level signature of $x|_{[s,t]}$ and is written as $S_N(x)_{s,t}$. In other words,

$$\begin{aligned} S_N(x)_{s,t} &\equiv \left(1, \int_{s < u < t} dx_u, \dots, \int_{s < u_1 < \dots < u_k < t} dx_{u_1} \otimes \dots \otimes dx_{u_k} \right) \\ &\in \oplus_{k=0}^N \left(\mathbb{R}^d \right)^{\otimes k}, \end{aligned}$$

Then $S_N(x)_s, \cdot$ satisfies the controlled DE by x as follows

$$\begin{cases} dS_N(x)_{s,t} = S_N(x)_{s,t} \otimes dx_t, \\ S_N(x)_{s,s} = 1. \end{cases}$$

Rough differential equation

$$dy = V(y)dx \Rightarrow y_{s,t} \approx \tilde{V}(y_s) \otimes S_N(x)_{s,t}$$